

From Measurement to Decision

An AI strategy for carbon-aware product design and procurement

How NVIDIA can evolve AI-assisted product carbon footprinting into predictive, decision-oriented carbon intelligence.

MEASURE

What
happened?

PREDICT

What will
happen?

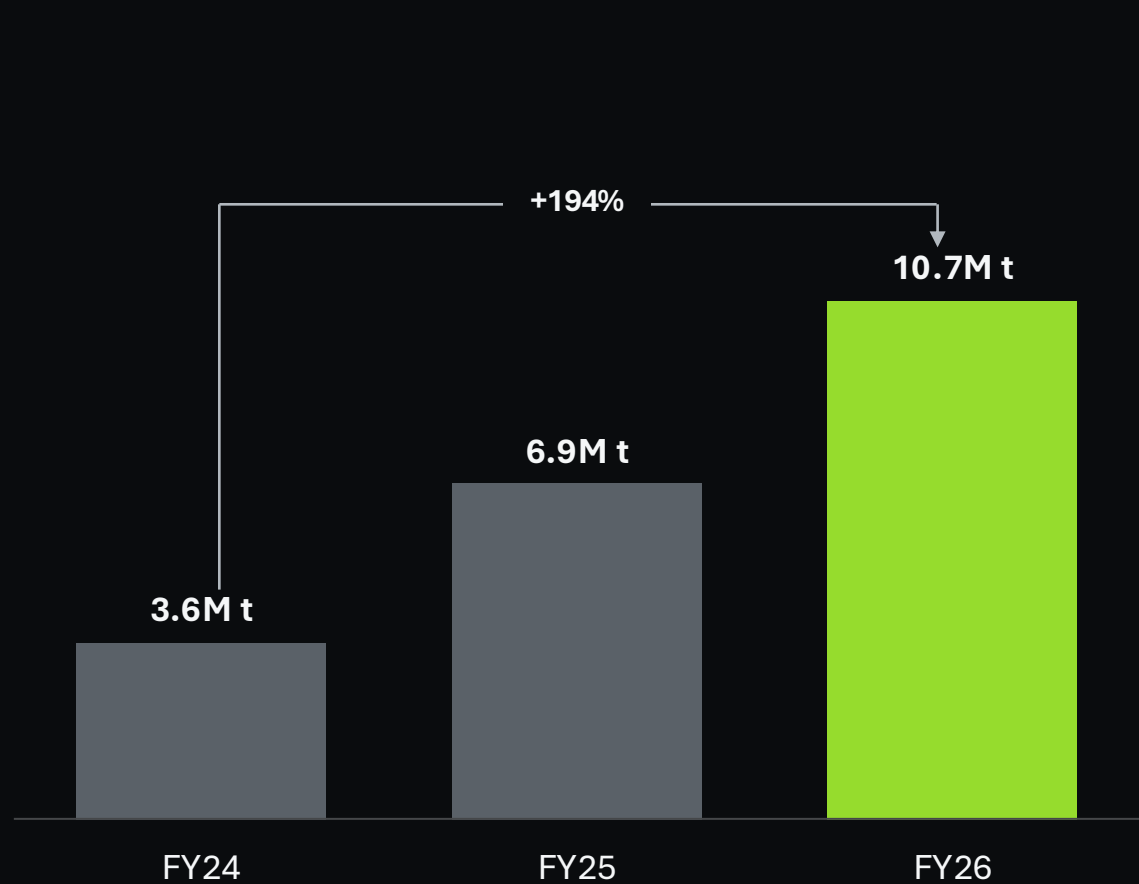
DECIDE

What should
we do?

Scope 3 emissions nearly tripled to 10.7M t CO₂e by FY26

Most of NVIDIA's footprint is created in its supply chain, not in its own operations.

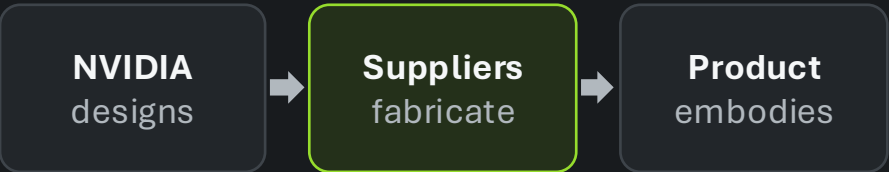
Scope 3 emissions, M t CO₂e



87%

of Scope 3 is **purchased goods & services**
(9.3M t CO₂e)

Where the carbon is created

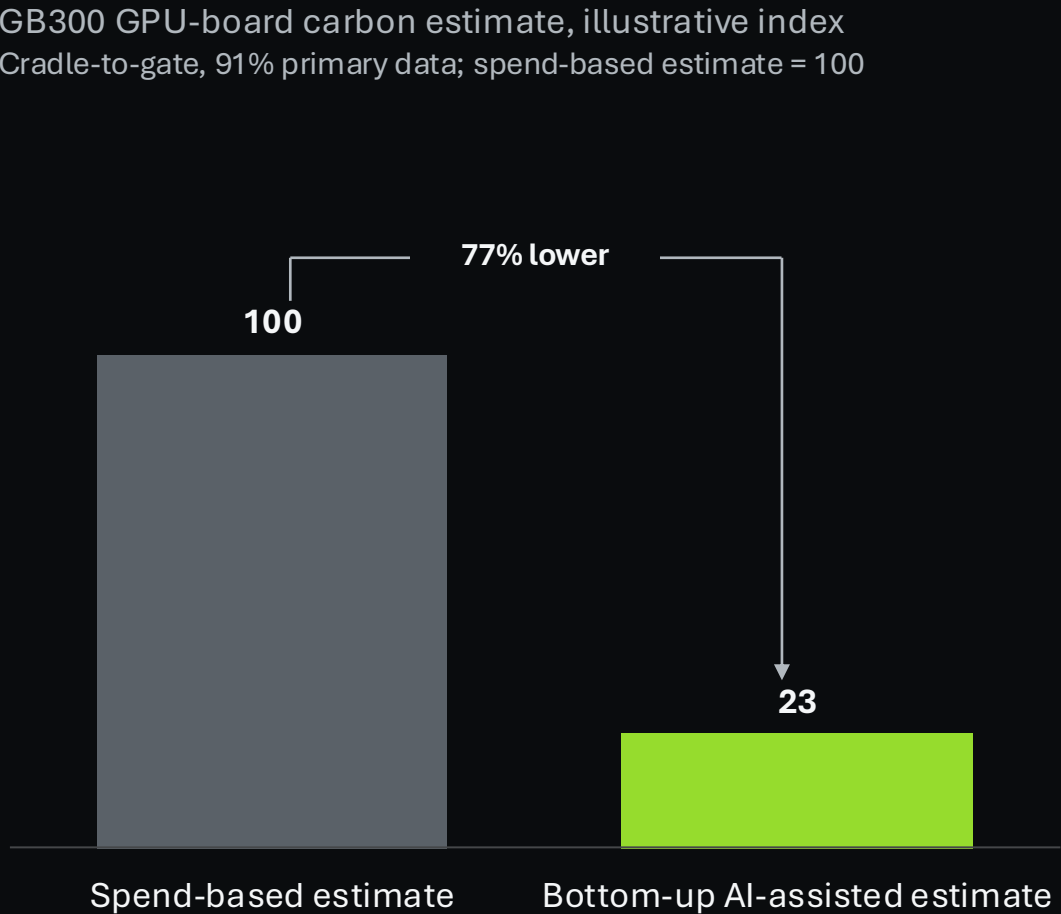


Silicon, memory, PCB and packaging plants emit the carbon NVIDIA's designs specify.

Source: NVIDIA FY26 Sustainability Report, Sustainability Indicators; product carbon footprint disclosures.

Bottom-up GB300 estimate was 77% below spend-based

Generic emission factors overstate the carbon of NVIDIA’s most advanced hardware.



Data quality, from preferred to fallback

- 1 Supplier primary data**
Actual production energy, materials, process
- 2 Semiconductor models**
Process node, die area, fab location, packaging
- 3 Generic lifecycle factors**
Fallback; misses advanced-node realities

More demanding than spend-based or parametric methods, but **the only one that resolves carbon at the component level**, where design decisions are made.

Records the system already processes

50K+
BOM records

12K+
ICs

250K+
substances

Illustrative index: $100 \times (1 - 0.77) = 23$, derived from NVIDIA’s reported 77% difference; NVIDIA did not publish index values. Shows measurement accuracy, not an emissions reduction.
Source: NVIDIA, “NVIDIA Turns Product Carbon Data Into Actionable Lifecycle Insights With AI”; NVIDIA PCF methodology.

Today carbon is measured after design choices lock in

NVIDIA's stated next step: bring lifecycle metrics into materials design and procurement.

TODAY Measure: AI-assisted product carbon footprinting



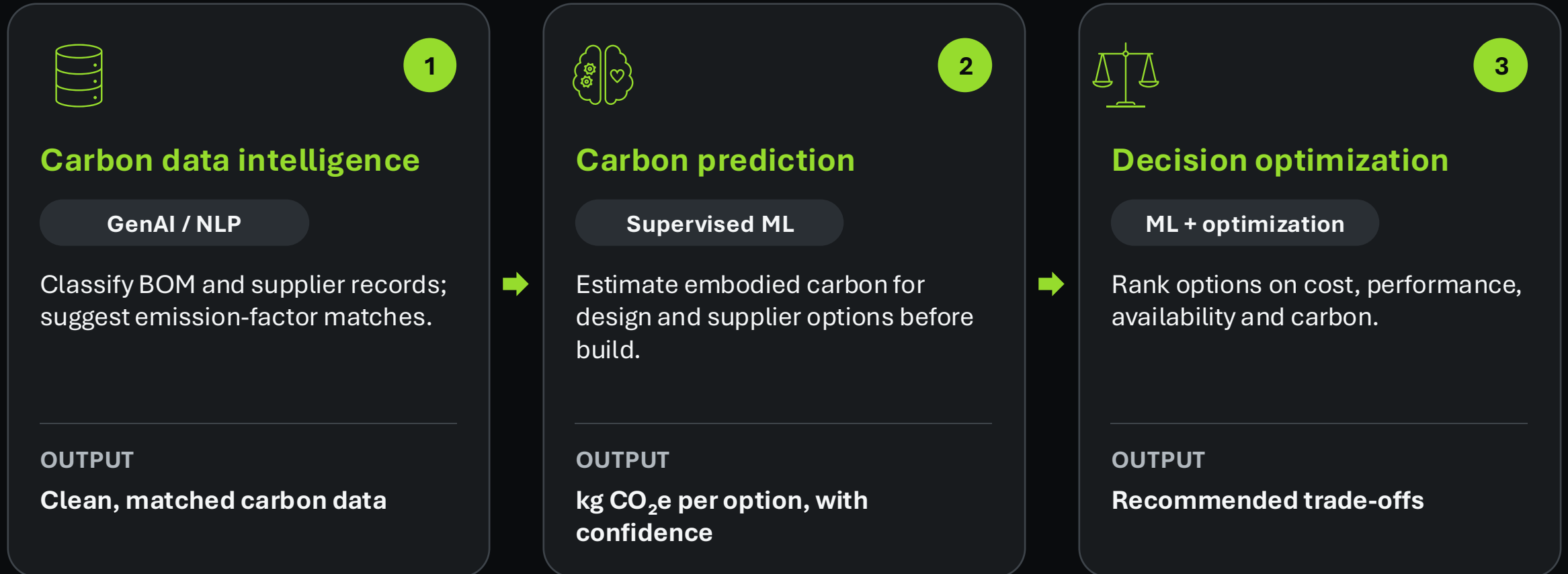
The shift: use the same data to inform choices **before** they are locked in, not to report after

PROPOSED Predict and decide: carbon-aware engineering and procurement



Three AI use cases turn carbon data into decisions

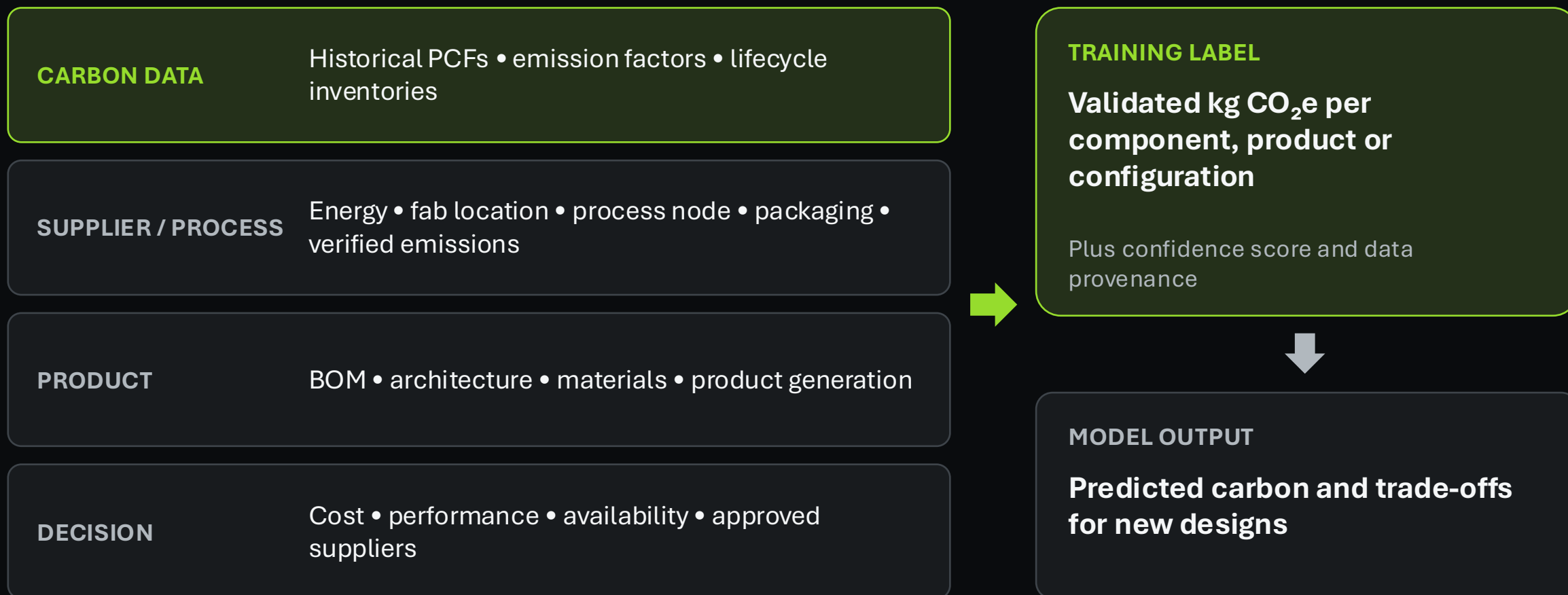
Each feeds the next: clean data trains prediction, and prediction drives optimization.



Next option: agentic monitoring flags supplier or design changes; people approve every sourcing and design decision.

Training labels come from NVIDIA's validated footprints

Every completed product footprint becomes a labeled example for the prediction model.



Build what differentiates NVIDIA; buy standard carbon data

Proprietary product and supplier context is the advantage; emission-factor data is not.

BUILD / CUSTOMIZE

Owned by NVIDIA

- Predictive carbon models
- Multi-objective optimization
- Integration with product workflows
- Supplier and component intelligence
- Internal decision interface

Why: Uses proprietary design and supplier data competitors cannot copy.

BUY / PARTNER

Vendor systems with “AI inside”

- Lifecycle inventory databases
- External emission-factor datasets
- Industry benchmark datasets
- Verification and assurance support

Why: Standard carbon data is costly to recreate and not a differentiator.

A six-role team pairs carbon experts with AI builders

This is not only a data-science problem: the people who decide must trust and use the outputs.

BUSINESS AND DOMAIN



AI Product Lead

Translator: owns use case, roadmap, adoption



Sustainability / LCA

Carbon methodology and validation



Design & Sourcing

Engineers and buyers act on outputs

TECHNICAL



ML / Data Science

Prediction and optimization models



Data Engineering

Pipelines, lineage and data quality



Responsible AI

Controls, auditability and risk

Critical translator role: connects carbon science, AI, engineering and business trade-offs.

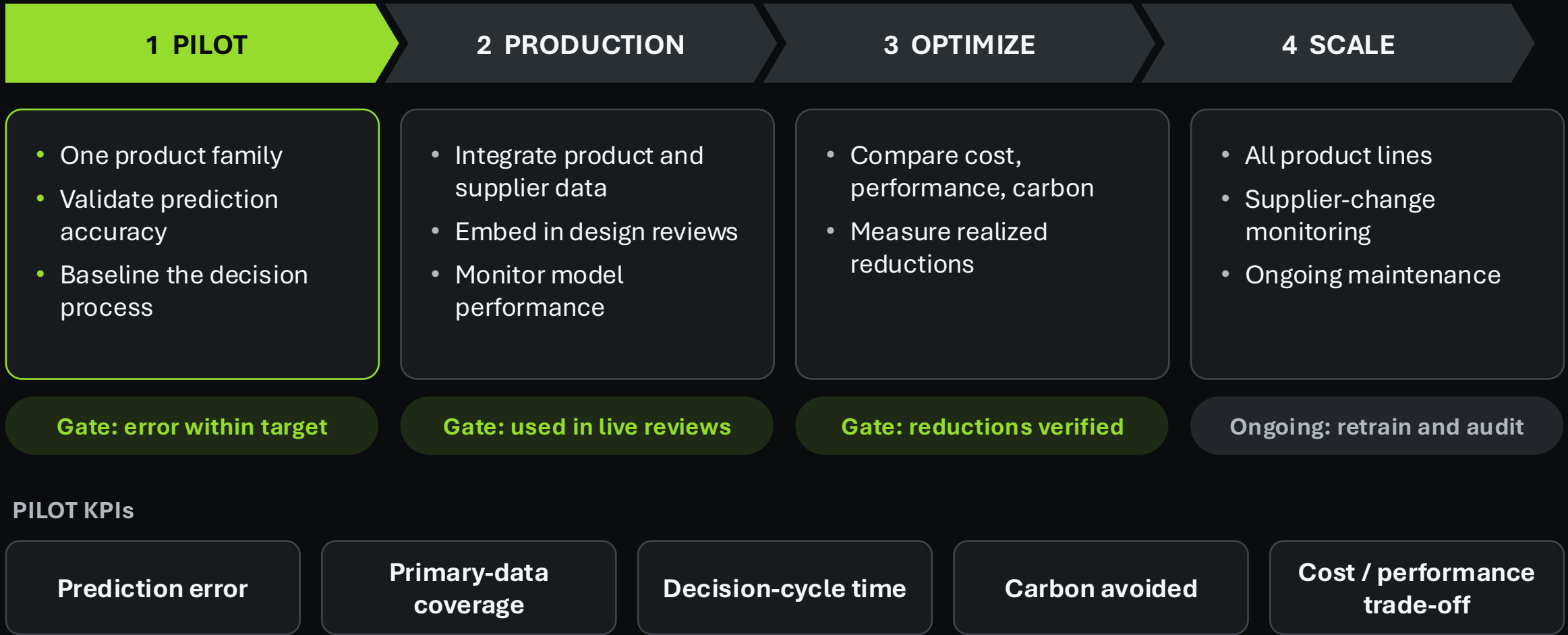
AI recommends; people decide every carbon trade-off

No supplier or design change happens solely because the model predicts lower carbon.



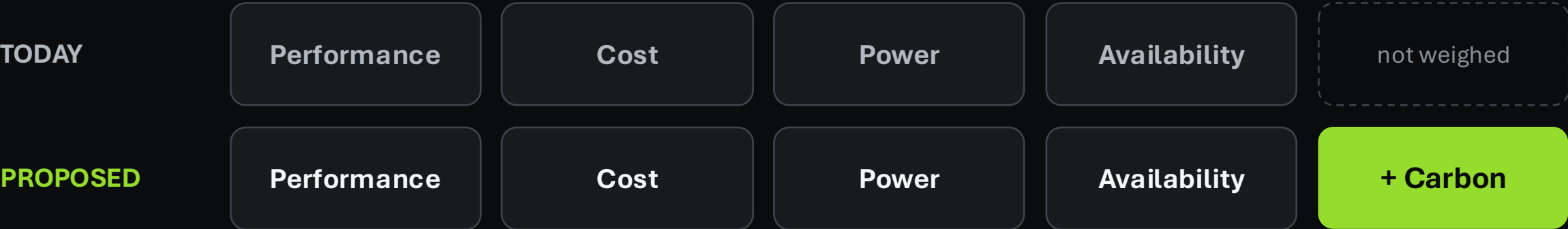
Pilot one product family, then scale on proven accuracy

Each phase passes an exit gate before the next; maintenance continues after scale.

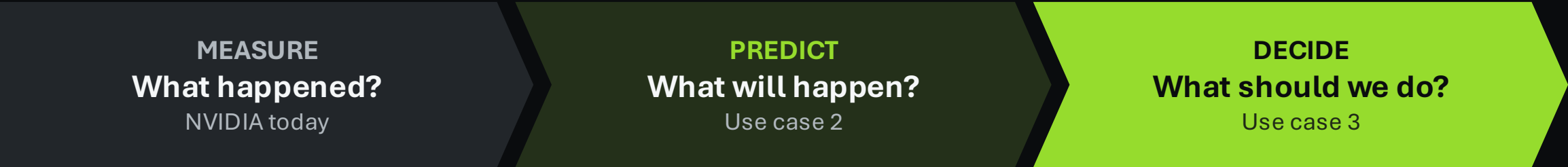


Carbon becomes a fifth engineering variable

The differentiator is better product and sourcing decisions, not better reporting.



FROM CARBON ACCOUNTING TO CARBON-AWARE DECISIONS



If NVIDIA surfaces credible carbon trade-offs early, **sustainability becomes part of product architecture**, not an after-the-fact report.

Proposed strategic differentiation; not a claim that NVIDIA currently operates this decision system.

Sources and assignment alignment

SOURCES

NVIDIA. FY26 Sustainability Report. 2026.

NVIDIA. “NVIDIA Turns Product Carbon Data Into Actionable Lifecycle Insights With AI.”

NVIDIA. Corporate Sustainability / Product Carbon Footprint resources.

CO2 AI. “How AI Is Revolutionizing Corporate Climate Action.”

Davenport, T. H. Developing an AI Strategy and Use Cases. Babson College, Sept. 23, 2025.

All the sources are verified

Appendix: AI prompts used

Prompts used with an AI assistant to research NVIDIA and build this presentation, in order.

1 “I would like to choose NVIDIA. Can I develop my AI strategy presentation around NVIDIA?”

2 “Is there data showing NVIDIA’s Scope 3 emissions? What exactly is NVIDIA’s Scope 3 challenge, how is it related to its supply chain, and what is NVIDIA currently doing about it?”

3 “Is NVIDIA currently using AI to analyze its bill of materials (BOM) and identify the components and substances contributing to a product’s carbon footprint? What does NVIDIA do with this information after identifying the carbon hotspots?”

4 “How does NVIDIA’s AI calculate or help estimate the carbon footprint associated with components in its bill of materials? Does the calculation incorporate supplier-specific information, such as manufacturing processes, energy sources, and carbon-reduction initiatives?”

5 “If different suppliers manufacture similar components but have different energy sources or carbon-mitigation programs, how does NVIDIA account for those differences? How are supplier-specific data and generic emission factors used in the carbon-footprint calculation?”

6 “Using the professor’s AI strategy framework and everything we have discussed about NVIDIA’s existing carbon-footprinting capability, create the presentation and organize it into the required slides.”

Prompts 1–5 gathered the NVIDIA research; prompt 6 organized it into the framework’s required slides.